

FREEMAN YIU

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EDUCATION

California State Polytechnic University, Pomona
Bachelor of Science, Computer Science; GPA: 3.94

Pomona, CA
Expected December 2026

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Libraries & Frameworks: React, FastAPI, PyTorch, Scikit-learn, LangChain, NumPy, Pandas

Technologies: Linux, Git, GitHub, Docker, PostgreSQL, AWS

EXPERIENCE

General Motors Cruise

Sunnyvale, CA

AI/ML Engineer Intern, Subsystem Evaluation

May 2026 - August 2026

- Expanded simulation coverage across 3 AV layers by implementing evaluation metrics for hard braking, swerving, and other driving behaviors in Python and SQL
- Automated layer attribution across the AV stack through a Streamlit dashboard, reducing the time engineers spent tracing the source of issues during root cause analysis
- Reduced false positives in simulation evaluation by investigating inconsistent scenarios in Webviz and identifying faulty cases with simulation engineers

Cal Poly Pomona Northrop Grumman Collaboration Project

Pomona, CA

UGV Integration Software Engineer

August 2025 - May 2026

- Enabled end-to-end autonomous UGV operations by collaborating with a cross-functional team across software, mechanical, and electrical disciplines to align integration procedures
- Enhanced system consistency by deploying ROS 2 publisher-subscriber nodes on an Ubuntu Linux VM
- Reduced debugging time by 20% using ROS graph visualization, system logs, and UDP socket testing to resolve Ethernet communication issues between ROS 2 nodes and STM32 microcontrollers

Cal Poly Pomona Autonomous Vehicle Lab - UAV Project

Pomona, CA

Computer Vision Team Member

August 2025 - May 2026

- Improved perception accuracy with a PyTorch YOLOv8 object detection model trained on 11,000+ thermal images
- Performed error analysis through iterative experiments by tuning hyperparameters and analyzing metrics (mAP, precision/recall/F1 curves, confusion matrix), reducing overfitting issues during training
- Tackled class imbalances via data augmentation, improving metrics by 8% (mAP: 0.42 to 0.5)

PROJECTS

Diabetes Classifier | *Python, Pandas, Matplotlib, Scikit-learn*

- Processed and filtered ~250k medical records to ensure high-quality data for predictive modeling
- Conducted EDA, feature engineering, and model comparison to identify key predictors and extract insights
- Designed and benchmarked multiple classifiers for best accuracy by analyzing precision/recall, F1 metrics
- Architected and deployed an interactive Streamlit app enabling real-time predictions from user health inputs

Legal Document RAG System | *Python, LangChain, ChromaDB, FastAPI*

- Automated low-latency inference and validation steps by architecting an LLM-powered backend service, reducing manual lookup time, and improving response latency
- Designed an ingestion pipeline including PDF parsing, text chunking, embedding generation, and vector retrieval with ChromaDB to allow faster retrieval speed in context searching
- Designed backend FastAPI routes to expose document ingestion, semantic search, and question answering